STAT-S 650 Time Series Analysis

Assignment 5

Manish Suresh Maudgalya

Theory

A. Do the 5 estimated parameters b0, b1, b2, b3, √ MSE match the true parameters?

```
Comparing Estimated and True Values
> true_params <- c(b0 = 10, b1 = 0.75, b2 = -0.65, b3 = 0.15, sqrt_MSE = 5)
> est         <- coef(fit)
> mse         <- mean(fit$residuals^2)
> sqrt_mse    <- sqrt(mse)
>
> cat("b0","\nTrue = ",true_params["b0"],"\nEstimated = ",est["(Intercept)"])
b0
True =  10
Estimated =  9.611641> cat("b1","\nTrue = ",true_params["b1"],"\nEstimated = ",est["y_lag1"])
b1
True =  0.75
Estimated =  0.714602> cat("b2","\nTrue = ",true_params["b2"],"\nEstimated = ",est["y_lag2"])
b2
True =  -0.65
Estimated =  -0.6151205> cat("b3","\nTrue = ",true_params["b3"],"\nEstimated = ",est["x_lag1"])
b3
True =  0.15
Estimated =  0.1594356> cat("sqrt(MSE)","\nTrue = ", true_params["sqrt_MSE"], "\nEstimated = ",sqrt_mse)
sqrt(MSE)
True =  5
Estimated =  4.839636>
```

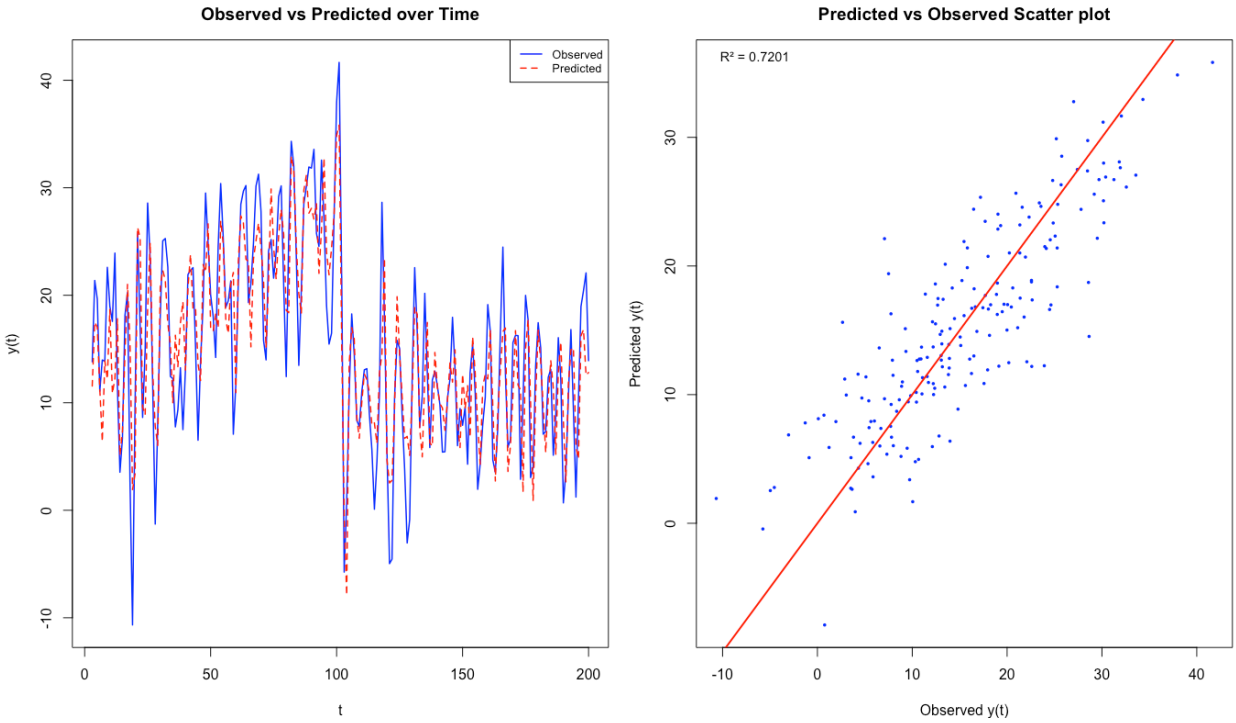
Looks like the Estimated Values are very Close to the True Values. Since we know the regressors are predetermined, OLS here is consistent.

B. What is the R^2 for the model?

```
>
> # R-squared
> r2 <- summary(fit)$r.squared
> cat("R-squared = \n", r2)
R-squared =
 0.7201229>
>
```

The R-Squared value is 0.7201229.

C. Plot the predicted values from the model against the observed values. Do the predictions match the observed very well?



As we can see in the line chart that the estimated values follow the observed lines quite closely. In the Scatter plot, the predicted and observed values also quite close to each other.

Practical

A. **Pick at least one output variable and at least one input variable from your data set. Formulate at least two models, a simple and a complex, with the simple being nested within the complex. Formulate a third model that is not nested within the other two models.**
B. **Fit the models using a linear regression program and assume white noise inputs.**

   All three models were fit using OLS assuming white noise errors, with a R^2 of 0.9045 for mode 10.9047 for model 2 and 0.6450 for model 3.

C. **Compare the complex and nested models using an F test**
   The F-test (F = 4.08, p = 0.017) rejects the null hypothesis that the lag terms are jointly zero, meaning Model 2 is statistically better than Model 1.

D. **Compare the non-nested model to the other two using BIC and AIC.**
   AIC and BIC: Model 2 has the lowest AIC −34330.94, but BIC is better in Model 1 −34307.88 vs −34299.45 due to its more complexity penalty, and Model 3 is bad on both criteria (AIC = −29045, BIC = −29013).
E. **Choose the best model of the three you compared. Justify your choice.**
   Model 2 is the best overall because it wins on AIC and also by the F-test, and the marginal BIC difference from Model 1 is small while the lag terms do add statistically significant explanation.

F. **Examine the residuals of the model to see if they are white noise using autocovariance on the residuals and spectral analysis of the residuals.**
   The Ljung-Box test ($\chi^2 = 76.75$, $p = 1.39\times10^{-8}$) strongly rejects the null that residuals are white noise, indicating significant remaining autocorrelation in Model 2's residuals.

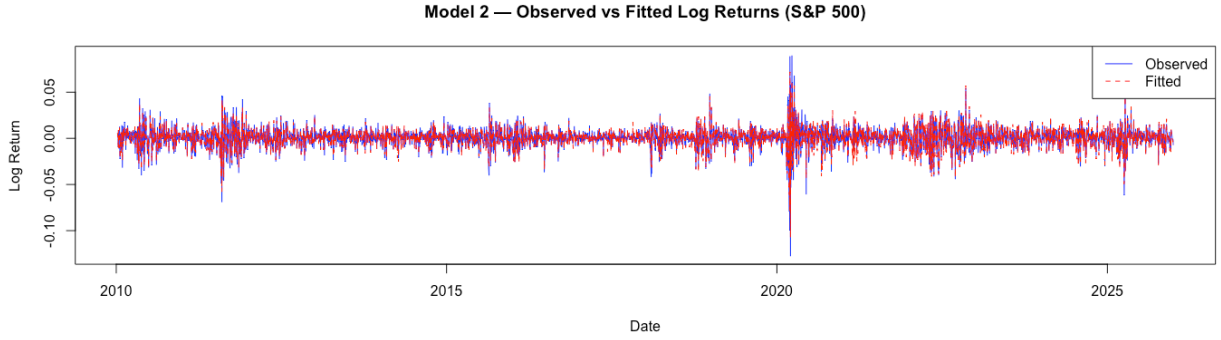
G. **If the residuals are not white noise, add more output feedback terms (increase the lag p of the output feedback) and see if this reduces the residuals to white noise.**

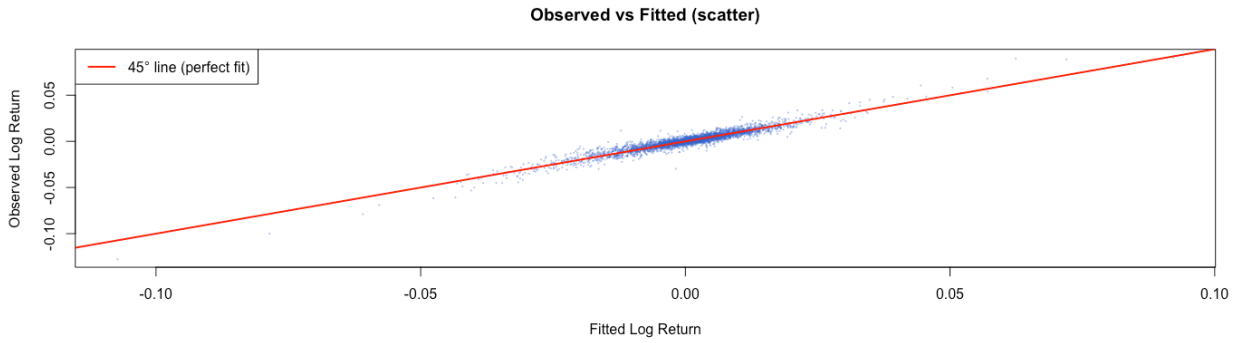
   Adding a third lag (Model 2 extended) did not resolve it. The Ljung-Box test remained highly significant and both AIC and BIC became worse, so increasing lag order is not enough to make the residuals white noise.

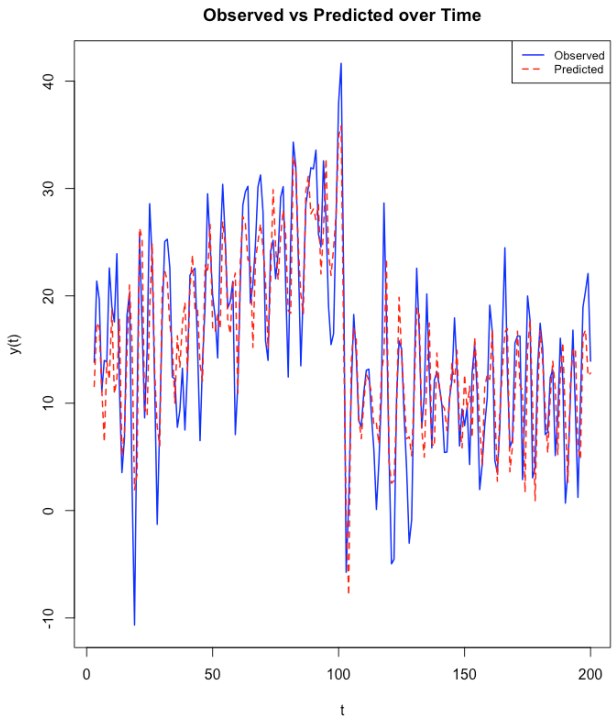
H. **Interpret the parameters of the model. What do they imply for the effect of the lagged feedback? Is the model stable? What do they imply for the effect of the input on the output?**

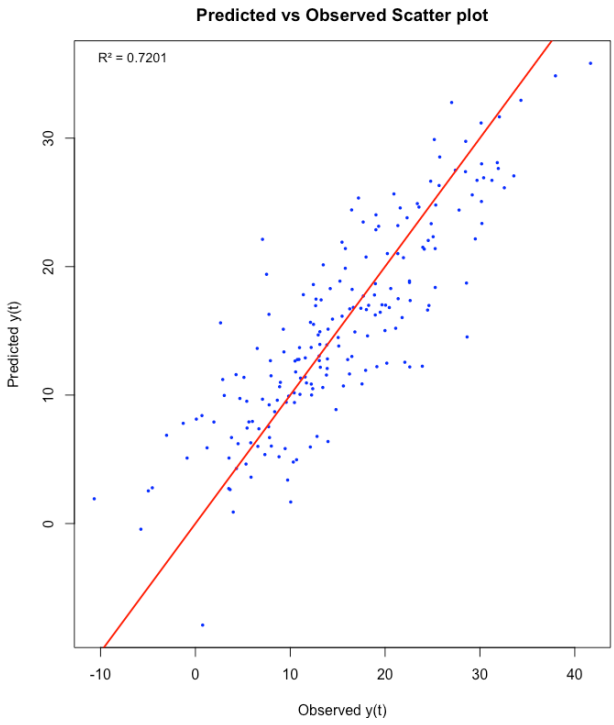
   The NASDAQ coefficient of 0.802 means a 1% NASDAQ return gives a 0.80% S&P 500 return on the same day. The model is not stable**,** roots have moduli of 0.097 and 0.106, both below 1, not quite stable.

Plot the model predictions for the output and compare these to the observed.









R Code

```r
set.seed(42)
T <- 200
#Generate x(t)
x <- ifelse(1:T <= 100, 1:T, 0)
#Generate y(t)
w <- rnorm(T, mean = 0, sd = 1)

y <- numeric(T)
y[1] <- 2
y[2] <- 4

for (t in 3:T) {
  y[t] <- 10 + 0.75 * y[t-1] - 0.65 * y[t-2] + 0.15 * x[t-1] + 5 * w[t]
}

# Fitting the model for t > 2:
#y(t) = b0 + b1*y(t-1) + b2*y(t-2) + b3*x(t-1) + error
# For values t = 3 to 200
t_idx <- 3:T
df <- data.frame(y_t   = y[t_idx],y_lag1 = y[t_idx - 1],y_lag2 = y[t_idx - 2],
                 x_lag1 = x[t_idx - 1]
)

fit <- lm(y_t ~ y_lag1 + y_lag2 + x_lag1, data = df)
summary(fit)


# a) Comparing estimated vs true parameters

cat("Comparing Estimated and True Values \n")
true_params <- c(b0 = 10, b1 = 0.75, b2 = -0.65, b3 = 0.15, sqrt_MSE = 5)
est         <- coef(fit)
mse         <- mean(fit$residuals^2)
sqrt_mse    <- sqrt(mse)

cat("b0","\nTrue = ",true_params["b0"],"\nEstimated = ",est["(Intercept)"])
cat("b1","\nTrue = ",true_params["b1"],"\nEstimated = ",est["y_lag1"])
cat("b2","\nTrue = ",true_params["b2"],"\nEstimated = ",est["y_lag2"])
cat("b3","\nTrue = ",true_params["b3"],"\nEstimated = ",est["x_lag1"])
cat("sqrt(MSE)","\nTrue = ", true_params["sqrt_MSE"], "\nEstimated =
",sqrt_mse)


# R-squared
r2 <- summary(fit)$r.squared
cat("R-squared = \n", r2)


# Plot predicted vs observed

```

```r
y_hat <- fitted(fit)
y_obs <- df$y_t

par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))

# Plot 1: Time series overlay
plot(t_idx, y_obs, type = "l", col = "blue", lwd = 1.5,
     xlab = "t", ylab = "y(t)",
     main = "Observed vs Predicted over Time")
lines(t_idx, y_hat, col = "red", lwd = 1.5, lty = 2)
legend("topright", legend = c("Observed", "Predicted"),
       col = c("blue", "red"), lty = c(1, 2), lwd = 1.5, cex = 0.8)

# Plot 2: Scatter predicted vs observed
plot(y_obs, y_hat, pch = 16, cex = 0.5, col = "blue",
     xlab = "Observed y(t)", ylab = "Predicted y(t)",
     main = "Predicted vs Observed Scatter plot")
abline(0, 1, col = "red", lwd = 2)
legend("topleft", legend = sprintf("R² = %.4f", r2), bty = "n", cex = 0.9)

```

R Code

```r
#Load Data
data <- read.csv("/Users/manishsm/Desktop/Spring 2026/Time Series Analysis/Assignments/quantmod_dataset.csv")
data$Date <- as.Date(data$Date)


#Log Returns
# log return: r_t = log(P_t / P_{t-1})
data$ret_GSPC <- c(NA, diff(log(data$GSPC.Close)))
data$ret_IXIC <- c(NA, diff(log(data$IXIC.Close)))

# Also compute log returns for IXIC OHLC (used in Model 3)
data$ret_IXIC_Open <- c(NA, diff(log(data$IXIC.Open)))
data$ret_IXIC_High <- c(NA, diff(log(data$IXIC.High)))
data$ret_IXIC_Low  <- c(NA, diff(log(data$IXIC.Low)))

#2. Add Lagged Output Variables (ARX feedback terms)
data$ret_GSPC_lag1 <- c(NA, head(data$ret_GSPC, -1))
data$ret_GSPC_lag2 <- c(NA, NA, head(data$ret_GSPC, -2))
data$ret_GSPC_lag3 <- c(NA, NA, NA, head(data$ret_GSPC, -3))
data_clean <- na.omit(data)
cat("Rows after computing log returns and lags:", nrow(data_clean), "\n")
cat("Date range:", as.character(min(data_clean$Date)),
    "to", as.character(max(data_clean$Date)), "\n\n")

# Quick summary of log returns
cat("Log Return Summary\n")
cat(sprintf("GSPC log return: Mean=%.5f, SD=%.5f\n",
            mean(data_clean$ret_GSPC), sd(data_clean$ret_GSPC)))
cat(sprintf("IXIC log return: Mean=%.5f, SD=%.5f\n\n",
            mean(data_clean$ret_IXIC), sd(data_clean$ret_IXIC)))


#  Fit Models white noise / iid error assumed via OLS
model1 <- lm(ret_GSPC ~ ret_IXIC, data = data_clean)
model2 <- lm(ret_GSPC ~ ret_IXIC + ret_GSPC_lag1 + ret_GSPC_lag2,
             data = data_clean)
model3 <- lm(ret_GSPC ~ ret_IXIC_Open + ret_IXIC_High + ret_IXIC_Low,
             data = data_clean)


cat("MODEL 1 SUMMARY (Simple - nested)\n")
cat("  ret_GSPC ~ ret_IXIC\n")
print(summary(model1))


cat("MODEL 2 SUMMARY (Complex - contains Model 1)\n")
cat("  ret_GSPC ~ ret_IXIC + ret_GSPC_lag1 + ret_GSPC_lag2\n")
print(summary(model2))

```

```r

cat("MODEL 3 SUMMARY (Non-nested)\n")
cat("  ret_GSPC ~ ret_IXIC_Open + ret_IXIC_High + ret_IXIC_Low\n")
print(summary(model3))


#  4: F-Test — Simple (Model 1) vs Complex (Model 2)
# H0: lag1 and lag2 coefficients are jointly zero (Model 1 is sufficient)
# H1: at least one lag term adds explanatory power


cat("F-TEST — Model 1 vs Model 2\n")
ftest <- anova(model1, model2)
print(ftest)
cat("If p < 0.05 -> reject H0 -> Model 2 significantly better than Model 1\n\n")


#  5: AIC and BIC — Compare All Three Models
cat(" 5: AIC vs BIC \n")

aic_vals <- AIC(model1, model2, model3)
bic_vals <- BIC(model1, model2, model3)

comparison_table <- data.frame(
  Model   = c("Model1 (Simple)", "Model2 (Complex)", "Model3 (Non-nested)"),
  Params  = aic_vals$df,
  AIC     = round(aic_vals$AIC, 2),
  BIC     = round(bic_vals$BIC, 2),
  R2      = round(c(summary(model1)$r.squared,
                    summary(model2)$r.squared,
                    summary(model3)$r.squared), 4)
)
print(comparison_table)
cat("\nLower AIC/BIC = better fit penalized for complexity.\n\n")


#  6: Choose Best Model
cat(" 6: BEST MODEL SELECTION\n")
cat("Best by AIC:", comparison_table$Model[which.min(comparison_table$AIC)], "\n")
cat("Best by BIC:", comparison_table$Model[which.min(comparison_table$BIC)], "\n\n")
cat("Justification: The best model balances goodness-of-fit (R2) and\n")
cat("parsimony (AIC/BIC penalty). If Model 2 wins on AIC/BIC AND the\n")
cat("F-test is significant, it is the clear best choice.\n\n")

# Use Model 2 as best model going forward
best_model <- model2


#  7: Residual White Noise Diagnostics

residuals_best <- residuals(best_model)
```

```r


cat(" 7: WHITE NOISE CHECK ON RESIDUALS\n")
# Ljung-Box test
lb_test <- Box.test(residuals_best, lag = 20, type = "Ljung-Box")
cat("Ljung-Box Test (lag = 20):\n")
print(lb_test)
cat("H0: residuals are white noise\n")
cat("If p < 0.05 -> residuals are NOT white noise -> go to  8\n\n")

# Plots
par(mfrow = c(2, 2))

acf(residuals_best, lag.max = 40,
    main = "ACF of Residuals (Model 2, log returns)",
    xlab = "Lag", ylab = "Autocorrelation")

pacf(residuals_best, lag.max = 40,
     main = "PACF of Residuals (Model 2, log returns)",
     xlab = "Lag", ylab = "Partial Autocorrelation")

spectrum(residuals_best, method = "pgram", taper = 0.1,
         main = "Spectral Density of Residuals",
         xlab = "Frequency", ylab = "Spectral Density", log = "no")

plot(data_clean$Date, residuals_best, type = "l", col = "blue",
     main = "Residuals Over Time",
     xlab = "Date", ylab = "Residual")
abline(h = 0, col = "red", lty = 2)

par(mfrow = c(1, 1))


#  8: Extend to lag p=3 if residuals not white noise


cat(" 8: EXTENDED MODEL WITH lag p = 3\n")


model2_ext <- lm(ret_GSPC ~ ret_IXIC + ret_GSPC_lag1 +
                   ret_GSPC_lag2 + ret_GSPC_lag3,
                 data = data_clean)
print(summary(model2_ext))

residuals_ext <- residuals(model2_ext)
lb_ext <- Box.test(residuals_ext, lag = 20, type = "Ljung-Box")
cat("Ljung-Box Test on Extended Model (lag = 20):\n")
print(lb_ext)

cat("\nAIC/BIC: Model2 (p=2) vs Extended (p=3):\n")
ext_compare <- data.frame(
  Model = c("Model2 (lag p=2)", "Model2_ext (lag p=3)"),
  AIC   = round(c(AIC(best_model), AIC(model2_ext)), 2),
  BIC   = round(c(BIC(best_model), BIC(model2_ext)), 2)
```

```r
)
print(ext_compare)
cat("\n")


#  9: Interpret Parameters + Stability Check


cat(" 9: PARAMETER INTERPRETATION\n")


coef_tbl <- summary(best_model)$coefficients
print(round(coef_tbl, 6))
cat("(Intercept)    : Average S&P 500 log return when all predictors = 0\n")
cat("ret_IXIC       : Beta — 1% NASDAQ return => this % change in S&P 500\n")
cat("ret_GSPC_lag1  : Momentum/feedback from S&P 500 return yesterday\n")
cat("ret_GSPC_lag2  : Feedback from S&P 500 return two days ago\n\n")

# Stability: for ARX model y_t = a1*y_{t-1} + a2*y_{t-2} + input terms
# Roots of characteristic polynomial z^2 - a1*z - a2 must have |root| > 1
a1 <- coef(best_model)["ret_GSPC_lag1"]
a2 <- coef(best_model)["ret_GSPC_lag2"]

char_roots <- polyroot(c(-a2, -a1, 1))
cat("Characteristic polynomial roots:\n")
print(char_roots)
cat("Moduli:\n")
print(round(Mod(char_roots), 4))
cat("Model is stable if all moduli > 1\n")
cat("Stable?", all(Mod(char_roots) > 1), "\n\n")


#  10: Plot Predictions vs Observed
cat(" 10: PREDICTIONS vs OBSERVED\n")
fitted_vals <- fitted(best_model)
obs_vals    <- data_clean$ret_GSPC

rmse <- sqrt(mean((obs_vals - fitted_vals)^2))
cat(sprintf("RMSE: %.6f\n", rmse))
cat(sprintf("R-squared: %.4f\n\n", summary(best_model)$r.squared))

par(mfrow = c(2, 1))

# Log return comparison
plot(data_clean$Date, obs_vals, type = "l", col = "blue", lwd = 0.8,
     main = "Model 2 — Observed vs Fitted Log Returns (S&P 500)",
     xlab = "Date", ylab = "Log Return")
lines(data_clean$Date, fitted_vals, col = "red", lwd = 1, lty = 2)
legend("topright", legend = c("Observed", "Fitted"),
       col = c("blue", "red"), lty = c(1, 2))

# Scatter: observed vs fitted
plot(fitted_vals, obs_vals, pch = 16, cex = 0.3,
     col = rgb(0.2, 0.4, 0.8, 0.4),
```

```r
     main = "Observed vs Fitted (scatter)",
     xlab = "Fitted Log Return", ylab = "Observed Log Return")
abline(0, 1, col = "red", lwd = 2)
legend("topleft", legend = "45° line (perfect fit)", col = "red", lwd = 2)

par(mfrow = c(1, 1))
```